Clearance & Elimination Practice 1

Module 4 - Drug Elimination and Clearance · Daily practice sheet · No date

Name _____ Date _____

(1st problem, unnumbered)

Estimate the CrCl of a 36-year-old male who weighs 160 lb, S_{Cr} =0.8 mg/dL, and 68” tall.

a The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Calculate the half-life of the agent.

b The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the amount of the agent in the body when the concentration of drug in the plasma is 60 mg/L?

c The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the rate of elimination (mg/hr) when the concentration of drug in the plasma is 50 mg/L?

d The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the rate of elimination (mg/hr) when the concentration of drug in the plasma is 25 mg/L?

e The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. What is the expected plasma concentration 12 hours after a 700-mg intravenous bolus dose of the agent?

f The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the fraction of drug excreted in the urine?

g The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the excretion rate constant of the agent?

h The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the renal clearance of the agent?

i The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What is the metabolic (hepatic) clearance of the agent?

j The average clearance and volume of distribution of an antiepileptic agent in the adult patient population are 0.5 L/hr and 9 L, respectively. Urine was collected for 96 hours and 25 mg of unchanged the agent was recovered from the original 700-mg dose. What AUC would you expect from the 700-mg dose?